Department of Computer Science and Engineering

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch	**:** V Semester, B.E CSE
Course Code & Title	**:** CB3491-Cryptography and Cyber Security
Name of the Faculty member	**:** Mrs.K.Jeyakarthika, AP / CSE

Innovative Practice Description

- **Unit / Topic: Unit II / Advanced Encryption Standard**
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO 7
- **Activity Chosen: Jigsaw Learning**
- **Justification:**

AES is a critical symmetric encryption algorithm with multiple stages such as key expansion, substitution, shifting, mixing columns, and adding round keys. Each of these components is intricate, requiring a deep understanding. The Jigsaw method allows students to break down these complex stages, focus on one, master it, and then teach their peers, ensuring that all team members have a solid grasp of each stage. This division fosters a deeper understanding of the individual parts as well as the algorithm as a whole.

- **Time Allotted for the Activity:** 40 Minutes
- **Details of the Implementation:**
  - After the completion of **AES algorithm with a 128-bit key**, including the number of rounds, the encryption steps (SubBytes, ShiftRows, MixColumns, AddRoundKey), and the key expansion process
  - Students were divided into **7 groups** with 5 or 6 members each. Within each group, **one member was assigned as leader** and asked them to focus on the remaining AES key sizes (192-bit or 256-bit). This division allowed every student to concentrate on understanding a specific aspect of AES in detail. After the initial study within groups, leader from each group, specializing in either **192-bit or 256-bit AES**, was selected to form an "Expert Group" for that specific key size.
  - **The expert groups** understand the encryption algorithm, focusing on their assigned key size. They discussed the **specific challenges**, **security implications**, and **performance trade-offs** associated with their assigned key length. Once the expert groups had thoroughly analysed their specific AES key sizes, they were dissolved, and all students were re-grouped into their **original teams**. Now each member in the team will be an expert in any one application. They shared their ideas to all the other members in the group that shows in Figure 1, that helps each student to learn all the algorithm easily for all key sizes. This enabled the students to get insight knowledge in this topic.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	3	1	2	1	3	3	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**
	3	3	3	3	**3**
Justification for correlation	To Apply the basic Computer Science Engineering and mathematics Knowledge for performing Symmetric encryption.	To analyze various Symmetric encryption algorithms	To provide the solutions with basic Symmetric encryption mechanisms	To investigate the various aspects of cryptanalysis.	Use the virtual environment and crypt tool for performing symmetric encryption
	PO8	**PO9**	**PO12**		
	1	2	1		
	To complete the assignments/ tasks in an ethical way	To work effectively as an individual and team for encrypt and decrypt a message using Symmetric Encryption Algorithms	Ability to update the Symmetric Encryption algorithms to the current trends.		

- **Images / Screenshot of the practice:**


Figure 1: Students discussion with their team members

- **Reflective Critique:**
    - **Feedback of practice from students and other stakeholders:**
        - Students felt the activity is very interesting.
        - Most of the students felt much more confident in my understanding of AES after teaching it to their peers.
    - **Benefit of the practice:**
        - Teamwork: Students learn to collaborate effectively with peers, discussing ideas, sharing insights, and working towards a common goal.
        - Communication: Presenting their findings to their original group fosters public speaking and explanation skills.
    - : **Challenges faced in implementation**
        - Students felt that at sometimes it was difficult to explain complex concepts clearly, but it helped me learn to simplify my explanations.
        - Few students hesitated to take part in expert group discussions
        - Individual assessment/observation was difficult.

References:

- https://www.jigsaw.org/
- https://www.readingrockets.org/strategies/jigsaw
- https://www.youtube.com/watch?v=euhtXUgBEts
- https://www.teachhub.com/teaching-strategies/2016/10/the-jigsaw-method- teaching-strategy/